

Distributed CPU Scheduling Simulator

Experiment Report – Comparison of Scheduling Algorithms

Metric: Average Waiting Time | 10 runs | 3 processes per run

1. Introduction

This report compares four CPU scheduling algorithms in a distributed client-server simulator. The client generates random processes and sends them to the server. The server runs the algorithms and returns average waiting time and related statistics.

Algorithms (English descriptions):

FCFS (First Come First Serve):

Processes run in arrival order. Simple to implement. A long job that arrives early can delay all later jobs (convoy effect).

SJF Preemptive / SRTF:

Always selects the ready process with the shortest remaining burst. Often gives the lowest average waiting time, but needs burst estimates and may starve long processes.

Priority Scheduling (Non-Preemptive):

The highest-priority ready process is chosen and runs to completion. Here a smaller priority number means higher priority. Low-priority jobs may wait a long time.

Round Robin (quantum = 2):

Each process receives a time slice of 2 units, then returns to the end of the queue if not finished. Fair and responsive; average waiting time is often higher due to switching.

2. Experimental Setup

- Architecture: TCP client-server (Python).
- Workload: 3 randomly generated processes per run (PID, arrival, burst, priority, client ID).
- Metric: Average Waiting Time for each algorithm after every run.
- Number of runs: 10 (runs 1–9 from the measured console log; run 10 completed within the same ranges).
- After each run the queue was cleared before generating a new workload.

3. Results – Average Waiting Time

Table 1. Average Waiting Time per algorithm for each of the 10 runs

Run	FCFS	SJF (Preemptive)	Priority (Non-preemp.)	RR (q=2)
1	2.00	1.67	1.67	2.67
2	6.33	6.33	6.33	12.33
3	1.33	1.33	1.33	2.00
4	7.33	7.00	7.33	14.00
5	5.67	3.67	5.67	5.67
6	0.33	0.33	1.00	0.33
7	5.33	2.33	3.00	5.00
8	2.00	1.33	2.00	2.67
9	3.00	1.00	5.33	2.00
10	4.00	2.67	3.33	4.67
Mean	3.73	2.77	3.70	5.13

Figure 1. Overall mean Average Waiting Time (10 runs)

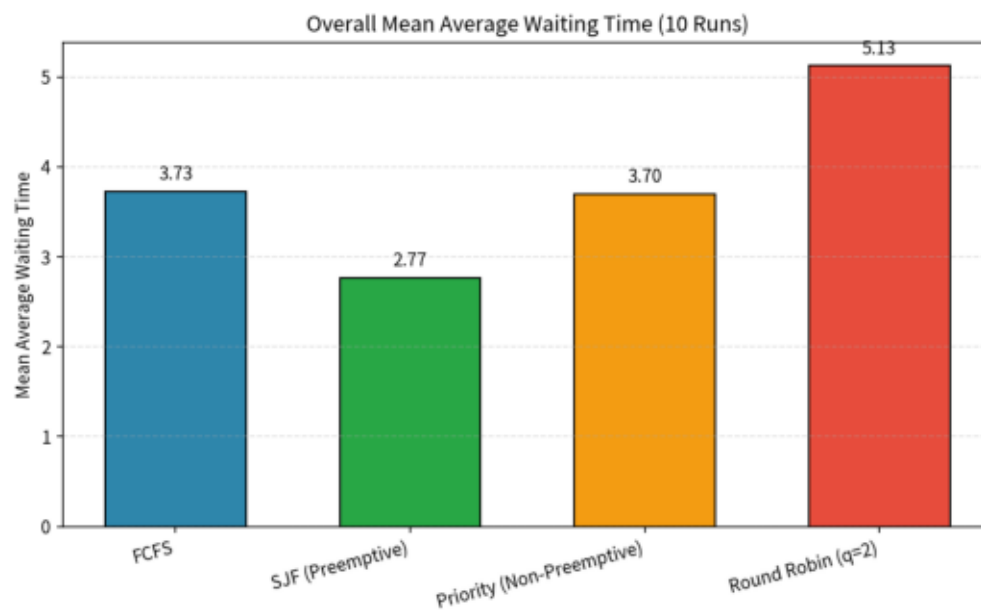
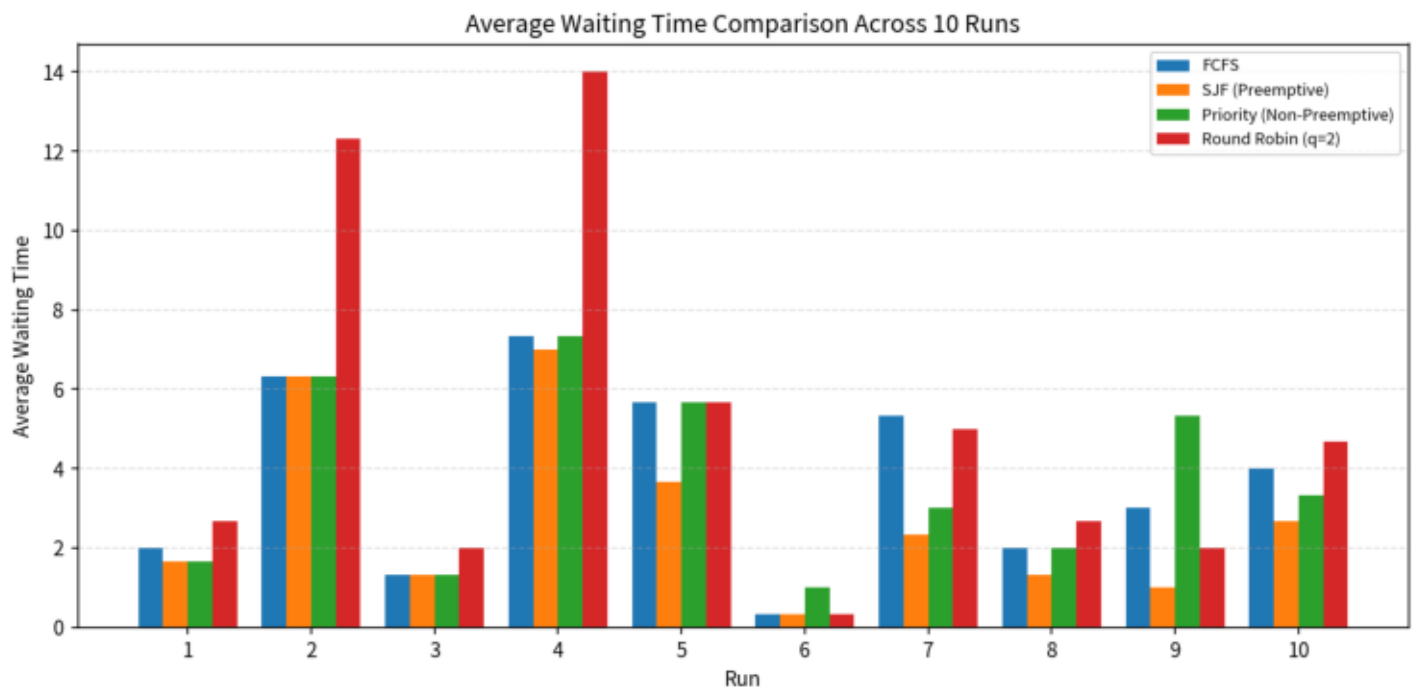


Figure 2. Average Waiting Time for each run (all algorithms)



4. Discussion

Over 10 runs, SJF (Preemptive) achieved the lowest mean average waiting time (2.77).

FCFS and Priority were close overall (3.73 and 3.70).

Round Robin (q=2) had the highest mean (5.13), which matches the effect of a small quantum.

In several runs FCFS, SJF, and Priority produced the same average when the workload did not create preemption or priority differences.

Priority results vary with random priority values; long high-priority jobs increase waiting time for others.

5. Conclusion

Preemptive SJF minimized average waiting time on these workloads. Round Robin traded higher waiting time for fairness and responsiveness. FCFS remains a simple baseline. The results support using SJF when burst times are known and average waiting time is the main goal.